

Technical Design Document

AI PR / Code Review Assistant — From Prototype to Production-Ready LLM Feature

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1. Problem & Business Value

1.1 The bottleneck

On most teams, opening a pull request is a tax on the most expensive people in the building. A developer finishes the actual work, then spends 15–30 minutes writing a description, reconstructing *why* each change was made, and self-reviewing against standards that live in a wiki nobody re-reads. Reviewers then re-derive the same context from scratch. The result is **slow, inconsistent PRs**: thin descriptions, missed standard violations, and review latency that blocks merges.

This is a high-value target because it sits on the critical path of *every* change a team ships, and the cost scales linearly with team size and velocity.

1.2 Target user & current workflow

- **Primary user:** the developer authoring a PR; **secondary:** the reviewer.
- **Current workflow:** finish code → manually write PR description → manually skim coding standards → open PR → reviewer reads diff cold → back-and-forth on things a checklist would have caught.

1.3 The proposed feature

Given a **git diff** and the **Jira ticket** that motivated it, the assistant produces:

1. a structured **PR description** (summary, changes, testing notes),
2. a **reviewer checklist** tailored to the diff and tied to team standards, and
3. **risk flags** with severity.

Crucially, the output is **grounded** in the team's own coding standards and related code via retrieval — it is not generic advice.

1.4 ROI

The blueprint's framing — "why spend \$0.05 per call?" — is worth inverting. If a call saved even 10 minutes of senior-engineer time (~\$1+ of loaded cost) and improved review quality, \$0.05 would be trivially justified. **But this design drives the marginal cost to ~\$0** (free Groq tier + local embeddings), so the ROI question collapses entirely: the only cost is engineering time, and the payback is measured in minutes saved per PR across the whole team.

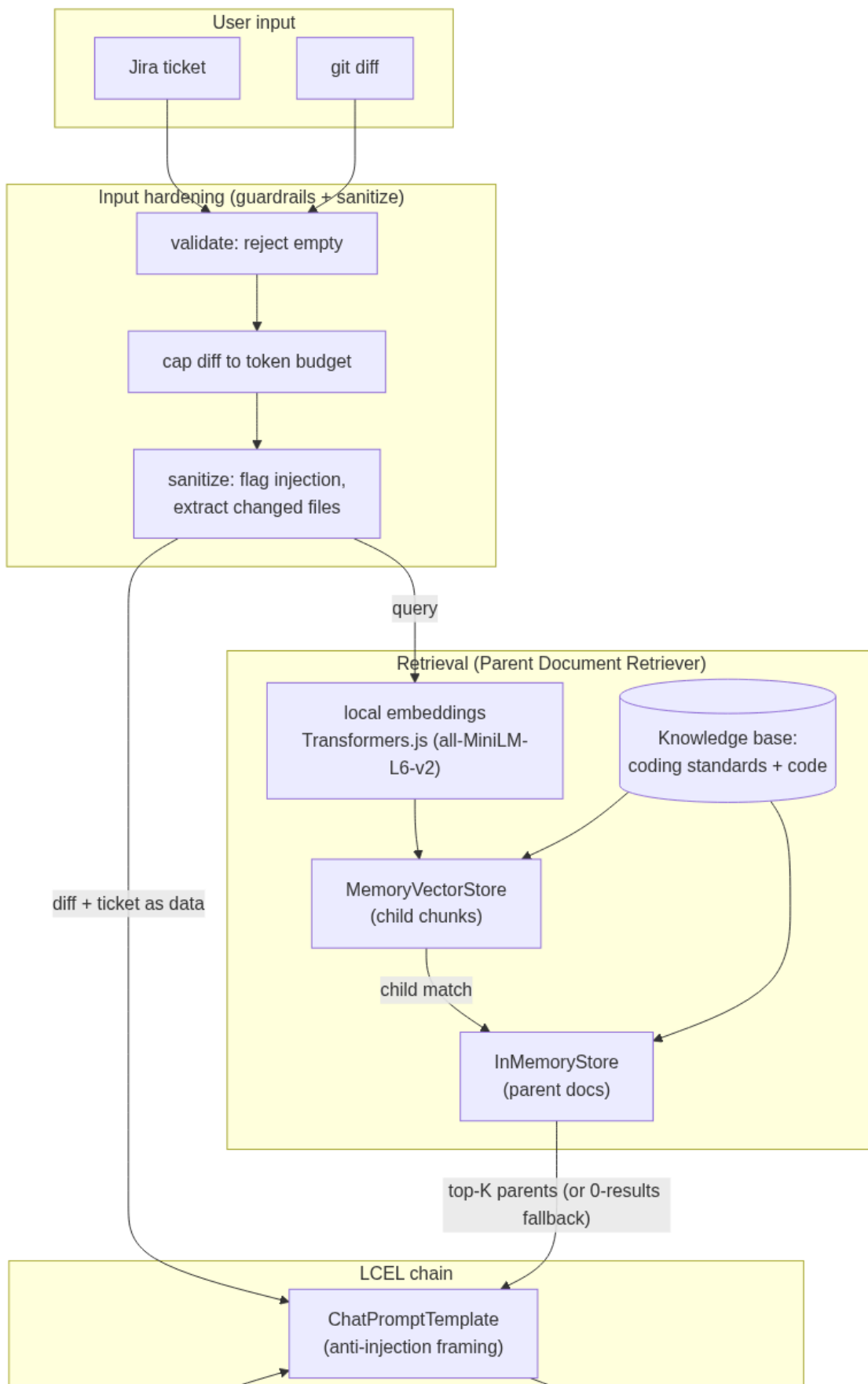
2. Use-Case Definition

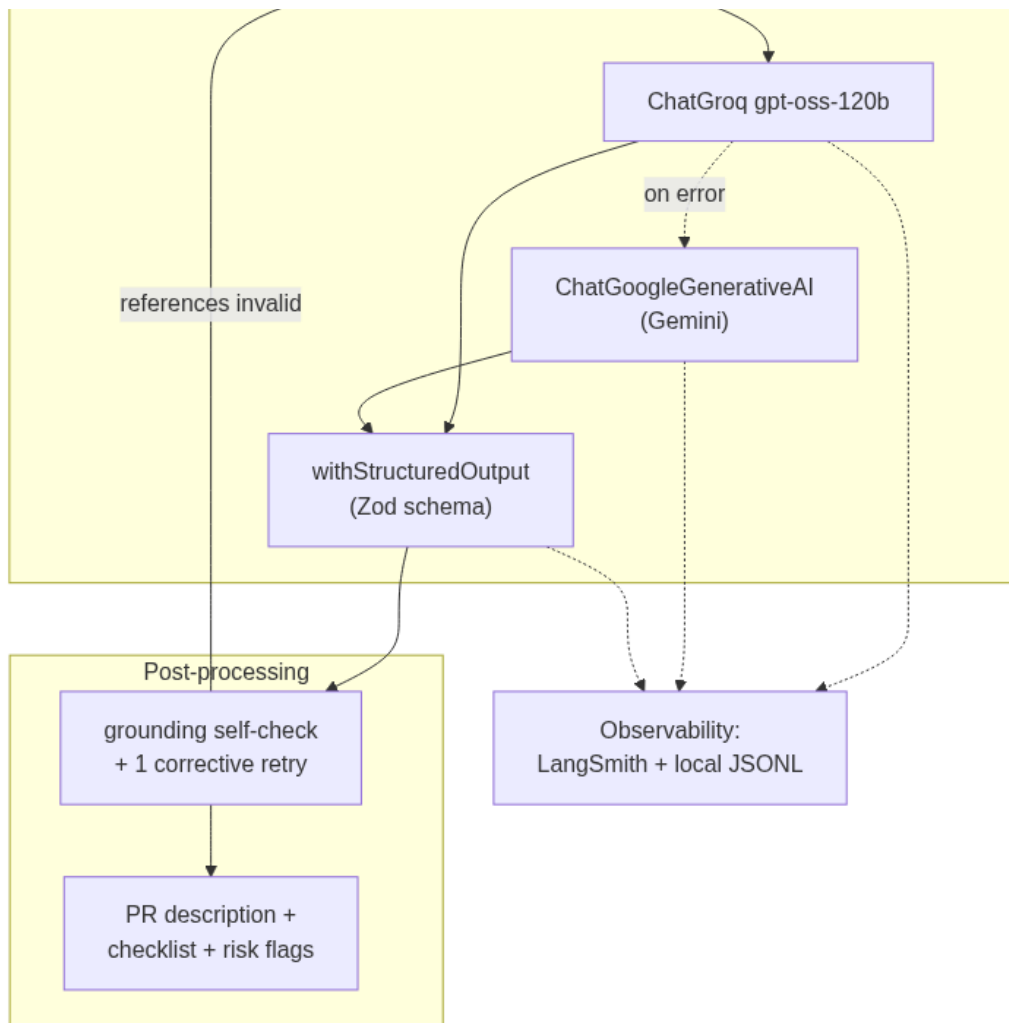
Dimension	Detail
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Trigger	Developer has a diff + ticket and is about to open a PR
Input	Unified git diff, Jira-style ticket (Markdown)
Output	PR description + reviewer checklist + risk flags (validated JSON)
Knowledge base	Team coding standards (6 docs) + a sample of existing code
Success	Output references only real files, cites relevant standards, catches the obvious risks, and never hallucinates a file path

3. System Architecture

The orchestration layer is built with **LCEL (LangChain Expression Language)**. The full data flow follows the required `User → Embedding → Vector Store → LLM → Output Parser` shape, with guardrails on either side:





3.1 Components

Component	Choice	Role
Embeddings	HuggingFaceTransformersEmbeddings (all-MiniLM-L6-v2)	Local, in-process, no API key
Vector store	MemoryVectorStore	Similarity search over child chunks
Doc store	InMemoryStore	Holds parent documents
Retriever	ParentDocumentRetriever	Advanced technique (see §4.3)
Prompt	ChatPromptTemplate	Anti-injection framing, strict grounding
LLM (primary)	ChatGroq — openai/gpt-oss-120b	Generation

LLM (fallback)	ChatGoogleGenerativeAI — gemini-2.5-flash	Resilience
Output parser	.withStructuredOutput(zodSchema)	Validated structured output
Observability	LangSmith (env-gated) + local JSONL	Tracing/evidence

3.2 Why LCEL over an agent

An agentic loop with tool-calling is the wrong tool here: the task is a **single, well-defined transformation** (diff + ticket + standards → structured review), not an open-ended goal requiring planning or external actions. LCEL gives a deterministic, traceable, low-latency pipeline with no risk of runaway tool loops. We keep one *bounded* feedback step — the grounding self-check — which is a deterministic guard, not an autonomous agent.

4. Key Technical Decisions ("the why")

4.1 Model choice — Groq GPT-OSS over gpt-4o-mini / Claude

The blueprint asks "why gpt-4o-mini over claude-3.5-sonnet?" The more interesting answer is **why neither hosted commercial model is the primary**:

Option	Cost (per ~1M tok)	Latency	Notes
Groq gpt-oss-120b	\$0 (free tier)	Very low (LPU)	Open-weight, data stays out of a vendor's training loop
OpenAI gpt-4o-mini	~\$0.15 in / \$0.60 out	Medium	Strong, cheap, but paid + closed
Anthropic Claude Sonnet	~\$3 in / \$15 out	Medium	Highest quality, ~20-100× the cost

For a **structured-extraction task** (the model fills a rigid Zod schema from provided context, it does not need frontier reasoning), `gpt-oss-120b` is more than sufficient — the evaluation below shows it producing accurate, grounded reviews. Groq's LPU inference makes it fast, and the free tier makes the per-call cost zero. **And because we built on LangChain, this is a one-line swap** — `LLM_PROVIDER=anthropic` — so the decision is reversible, not load-bearing.

4.2 Embeddings — local over hosted

Groq has no embeddings API, which forced a decision that turned into an advantage: **local embeddings via Transformers.js**. `all-MiniLM-L6-v2` runs in-process, needs no key, costs nothing, and keeps proprietary code from ever leaving the container. The weights are baked into the Docker image at build time, so retrieval works fully offline. Trade-off: lower embedding quality than `text-embedding-3-large` — acceptable for a small, well-structured corpus, and upgradeable by swapping one factory line.

4.3 Retrieval — Parent Document Retrieval over the alternatives

The required "Pro touch." For **code and standards**, a naive top-k chunk retriever is actively harmful: a 400-char chunk of a function or a standard is precise to *match* but useless as *context*. Parent Document Retrieval resolves this tension — embed **small child chunks** for precise matching, but return the **full parent document** to the LLM.

- *vs. naive RAG*: better context without sacrificing match precision.
- *vs. Self-Querying*: we kept metadata (`category` , `language`) on every doc so self-querying is a natural next step, but it adds an LLM call to parse the query — unjustified at this corpus size.
- *vs. Contextual Compression*: would save tokens, but with a free model and a small corpus the latency/complexity cost outweighs the benefit.

4.4 Structured output — `withStructuredOutput` + Zod

The output feeds a UI/automation, so it must be machine-parseable. Binding a Zod schema via `.withStructuredOutput()` collapses the "LLM → Output Parser" stages into one validated step. **Two real portability lessons surfaced during testing** (see §6.3): gpt-oss returns JSON as message *content* rather than a tool call (so we use `method: "jsonSchema"`), and Gemini's `responseSchema` rejects nullable type-unions (so the schema avoids `null / optional` in favor of required fields). The schema is now portable across both providers.

4.5 Cross-provider fallback — the point of LangChain

`ChatGroq(...).withFallbacks([ChatGoogleGenerativeAI(...)])` gives a **real, live failover** to a *different vendor*. This is LangChain's core value made concrete: the chain is written once and the vendor is a runtime detail. It is demonstrated live in §6.4, and it doubles as free-tier rate-limit insurance — when Groq returns 429, Gemini transparently answers.

5. Trade-off Analysis

Axis	Decision	Trade-off accepted
Cost vs. quality	Free open-weight model	Less raw capability than Claude — fine for structured extraction
Latency vs. accuracy	Parent Document Retrieval + self-check retry	Self-check can add one extra LLM call when grounding fails
Accuracy vs. context size	Return full parents	More tokens per call than bare chunks
Simplicity vs. power	LCEL chain, not an agent	No autonomous tool use (intentionally)

Throughput vs. cost	Free tier (8k tokens/min)	Calls must be paced; production would use a paid tier or queue
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6. Evaluation & Iteration

6.1 Methodology

A harness (`eval/run-eval.ts`) runs golden + failure cases, an A/B iteration comparison, and the live failover, writing JSONL traces + a Markdown report to `docs/evidence/` and streaming every run to LangSmith. **Final result: all checks passed.**

6.2 Edge-case results (from the live run)

Case	Expectation	Result
feature-discount-codes	grounded, no hallucinated files	☐ checklist=7, hallucinated=0, sources cited
bugfix-auth-expiry	grounded, cites security/auth standards	☐ checklist=6, sources=[security-guidelines, authMiddleware, ...]
empty-diff	rejected with <code>InputError</code> , no LLM call	☐ rejected before any model call
prompt-injection	flagged; not compromised	☐ injection flagged, checklist still produced (3 items)
oversized-diff	truncated to budget	☐ truncated=true
zero-retrieval	ungrounded fallback + note	☐ grounded=false, groundingNote set

6.3 Iteration — fixing a hallucination (the headline result)

The most instructive iteration was the grounding journey, captured by running the **same diff** through v1 and v2:

- **v1 (BEFORE):** a naive prompt with **no retrieval grounding**. The model invented **five file paths that do not exist in the diff**:
`src/db/migrations/XXXX_add_discount_to_orders.ts` , `src/app.ts` , `src/middleware/errorHandler.ts` , `tests/unit/orderService.test.ts` , `tests/integration/orders.test.ts` — a classic RAG-less hallucination.
- **v2 (AFTER): Parent Document Retrieval** + an explicit grounding clause (*"only reference files in the diff"*) + a deterministic **grounding self-check** that re-extracts file references from the output and triggers a corrective retry if any are invented.

	Hallucinated (out-of-diff) files
BEFORE (v1, ungrounded)	5 invented paths incl. <code>src/app.ts</code> , <code>src/middleware/errorHandler.ts</code> , <code>tests/unit/orderService.test.ts</code> (5)
AFTER (v2, grounded + self-check)	none (0)

This is the evaluation evidence: `docs/evidence/iteration-before.jsonl` vs `iteration-after.jsonl`, the `iteration-before / iteration-after` traces in `LangSmith`, and the `eval-report.md` summary.

A second, deterministic iteration came from output reliability: the first integration attempt returned `OUTPUT_PARSING_FAILURE` because `gpt-oss` emits JSON as content (not a tool call) and the schema rejected the model's `null`. Switching to `method: "jsonSchema"` and making the schema null-free/portable took parse success from 0% to 100% across **both** Groq and Gemini.

6.4 Live cross-provider failover

The harness builds a chain whose Groq primary uses a non-existent model id (guaranteed to 404), wrapped in a Gemini fallback. **Result: the primary failed (confirmed in isolation) and the chain still returned a valid, schema-validated review via Gemini** (`failover` check passed; `docs/evidence/failover.jsonl`). The failover is real, not theoretical.

7. Production Readiness

- **Monitoring:** `LangSmith` tracing is provider-agnostic and env-gated (`LANGSMITH_TRACING=true`); every run is labeled (`case-*` , `iteration-before/after` , `failover-groq-to-gemini`). A local JSONL logger guarantees evidence with zero external dependencies, recording provider chain, grounding, retrieval sources, truncation, injection flags, and retries.
 - **Security:** untrusted diff/ticket content is structurally framed as data (delimited `<untrusted_*>` blocks + an explicit system instruction), injection attempts are pattern-flagged and surfaced in traces (verified by the injection case), secrets live only in `.env` (gitignored), and embeddings run locally so code never leaves the host.
 - **Cost control:** \$0 marginal cost today; token caps (`MAX_DIFF_TOKENS`), and a documented path to caching/batching for scale.
 - **Rate limits & fallbacks:** the free tier caps ~8k tokens/min; the harness paces calls and the **Gemini fallback absorbs 429s**, alongside the zero-retrieval ungrounded path, empty-input rejection, and bounded retries.
 - **Scaling:** swap `MemoryVectorStore` for a persistent store (pgvector/Chroma) with incremental indexing; the retriever interface is unchanged. Run as a stateless container behind a queue for throughput.
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8. Limitations & Future Work

- In-memory vector store re-ingests per run — fine for this corpus, not for a large monorepo.
- Approximate (chars/4) token counting; production would use the model's tokenizer.
- The grounding self-check is heuristic (filename matching); a stronger version would use an LLM-as-judge grounding score.
- Self-Querying and Contextual Compression are designed-for but not enabled (see §4.3).

9. Conclusion

This is not "an API call." It is a modular, provider-agnostic RAG pipeline with a real advanced retrieval technique, structured and validated output, layered guardrails, live cross-provider resilience, and observability — packaged to run anywhere Docker runs, at zero marginal cost. The evaluation demonstrates accurate grounded reviews, correct edge-case handling, a measured hallucination fixed

(5 → 0 invented files), and a working cross-vendor failover. The design choices are deliberate, defended, and reversible.